

# Lecture 24: Final Project Work Session

## Models of Language and Conversation

PSYC 51.07: Models of Language and Communication

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# Today's Agenda

- 1 Final Project Overview
- 2 Presentation Guidelines
- 3 Technical Tips
- 4 Common Pitfalls
- 5 Grading & Expectations
- 6 Work Time & Support
- 7 Wrap-up

# Where We Are

## The Home Stretch

This is it—the final week of the course! Your final projects are due soon, and presentations are coming up.

### This week's schedule:

- **Today (Monday):** Project work session, debugging support, presentation prep
- **Wednesday:** Final project presentations (Part 1)
- **Friday:** Final project presentations (Part 2)
- **End of Friday:** All final materials due (11:59 PM Eastern)

## Important

Today is your **last structured work time** before presentations. Use it wisely!

# Final Project Requirements Review

## What you need to submit (all due Friday EOD):

- ① **Code Implementation:** Jupyter notebook that runs in Google Colab
  - Well-documented, reproducible, polished
  - Includes all experiments and results
- ② **Video Presentation:** 10-12 minute video (YouTube link)
  - Cover motivation, approach, results, discussion
  - All team members participate
- ③ **Written Writeup:** 2-5 page report
  - Intro, approach, results, discussion, references
  - Think mini research paper
- ④ **In-Class Presentation:** Wednesday or Friday
  - Show your video + lead 5-10 min Q&A discussion

# Presentation Format

## Video Component:

- **Length:** 10-12 minutes
- **Format:** Pre-recorded video
- **Platform:** YouTube (unlisted OK)
- **Content:**
  - Motivation (why?)
  - Approach (how?)
  - Results (what?)
  - Discussion (so what?)

## In-Class Component:

- **Show video** to class
- **5-10 min Q&A**/discussion
- **All team members** present
- Be ready for questions about:
  - Technical details
  - Design decisions
  - Limitations
  - Future work

### Pro Tip

Test your video beforehand! Make sure audio is clear, slides are readable, and demos work smoothly.

# What Makes a Great Presentation?

## Content:

- **Clear motivation:** Why does this problem matter?
- **Technical depth:** Show you understand the methods
- **Honest results:** Good and bad—what worked, what didn't?
- **Critical analysis:** Limitations, future work, broader implications

## Presentation quality:

- **Clear visuals:** Readable slides, effective figures
- **Good pacing:** Not too fast, not too slow
- **Engaging delivery:** Energy, clarity, enthusiasm
- **Team coordination:** Smooth transitions, balanced participation

### Think about it!

Your audience is smart but may not know your specific domain. Explain technical concepts clearly without oversimplifying.

# Presentation Structure Template

## Suggested 10-12 minute structure:

- ① **Title & Introduction** (1 min)
  - Project title, team members, one-sentence summary
- ② **Motivation & Background** (2 min)
  - The problem, why it matters, research questions
- ③ **Approach & Methods** (3-4 min)
  - Models, datasets, techniques, implementation details
- ④ **Results** (3-4 min)
  - Key findings, metrics, visualizations, demos
- ⑤ **Discussion & Conclusions** (1-2 min)
  - Limitations, future work, takeaways, broader impact

# Finishing Your Project: Technical Checklist

## Code & Implementation:

- ☐ Notebook runs completely in fresh Colab session (no errors!)
- ☐ All required packages installed with `!pip install` commands
- ☐ Dataset downloads automated (no manual file uploads)
- ☐ Random seeds set for reproducibility
- ☐ Code is well-commented and organized
- ☐ Markdown cells explain what's happening and why

## Results & Analysis:

- ☐ All experiments completed and documented
- ☐ Visualizations clear and informative
- ☐ Metrics and baselines included
- ☐ Results interpreted thoughtfully



# Testing Reproducibility

## Critical Step: Test in Clean Environment

Before submitting, open a **brand new** Colab session and run your notebook from top to bottom. This is what graders will do!

### Common reproducibility issues:

- 1 **Missing installations:** Did you forget a `!pip install`?
- 2 **File paths:** Are you referencing files that only exist on your machine?
- 3 **API keys:** Did you hard-code keys? (Use environment variables or instructions)
- 4 **Cell order:** Does notebook work when run sequentially?
- 5 **Out-of-order execution:** Did you run cells out of order during development?
- 6 **Runtime requirements:** Does it need a GPU? Does it exceed Colab limits?

### Pro Tip

Use "Runtime → Restart and run all" frequently during development!

# Code Quality Matters

## Good code is:

- **Readable:** Clear variable names, logical organization
- **Documented:** Comments, docstrings, markdown explanations
- **Modular:** Functions for repeated tasks, not copy-paste code
- **Efficient:** Reasonable runtime, not wasteful

## Example of good documentation:

```
def compute_embeddings(texts, model_name="sentence-transformers/all-  
MiniLM-L6-v2") :  
    """  
    Compute semantic embeddings for a list of texts.  
  
    Args:  
        texts (list): List of text strings to embed  
        model_name (str): HuggingFace model identifier  
  
    Returns:
```

# Common Pitfalls to Avoid

## Technical pitfalls:

- **Scope creep:** Trying to do too much at the last minute
- **No baselines:** Results without comparison are hard to interpret
- **Cherry-picking:** Only showing best results, hiding failures
- **Ignoring errors:** Sweeping bugs under the rug instead of fixing
- **Last-minute testing:** Discovering critical bugs right before deadline

## Presentation pitfalls:

- **Too much detail:** Getting lost in implementation minutiae
- **No motivation:** Jumping straight to methods without context
- **Unclear visuals:** Tiny fonts, cluttered slides, confusing plots
- **Going over time:** Practice and time yourself!
- **Unbalanced team:** One person doing all the talking

# When Things Don't Work

## Research Reality

Not everything will work perfectly. That's **completely normal** in research!

**If experiments fail or results are negative:**

- **Don't panic!** Negative results are still valuable
- **Analyze why:** What went wrong? What did you learn?
- **Be honest:** Discuss failures and limitations openly
- **Show process:** Document what you tried and why
- **Suggest improvements:** What would you do differently?

### Pro Tip

Projects are graded on process, rigor, and critical thinking—not just positive results. A thoughtful analysis of why something didn't work can be just as valuable as showing success!

# Time Management: The Final 72 Hours

## Recommended priorities:

- ① **Priority 1 (Critical):** Make code run and reproduce core results
  - Fix any breaking bugs
  - Ensure reproducibility
  - Complete essential experiments
- ② **Priority 2 (High):** Create presentation video
  - Draft slides
  - Record and edit video
  - Upload to YouTube
- ③ **Priority 3 (Medium):** Polish notebook
  - Add documentation
  - Clean up code
  - Add visualizations
- ④ **Priority 4 (Nice-to-have):** Written writeup refinement
  - Draft should be done, now refine
  - Proofread, fix formatting

# Grading Breakdown

## How your project will be evaluated:

| Component                       | Weight      |
|---------------------------------|-------------|
| Implementation (Code)           | 40%         |
| Results & Analysis              | 20%         |
| Presentation (Video + In-Class) | 15%         |
| Written Writeup                 | 10%         |
| Proposal                        | 10%         |
| Teamwork & Collaboration        | 5%          |
| <b>Total</b>                    | <b>100%</b> |

### Pro Tip

Focus most of your remaining time on **implementation** and **presentation**—they make up 55% of the grade!

# What We're Looking For

## Excellent projects demonstrate:

- **Technical sophistication:** Mastery of course concepts
- **Rigor:** Well-designed experiments, appropriate evaluation
- **Critical thinking:** Thoughtful analysis of results and limitations
- **Clarity:** Clear communication in code, writing, and presentation
- **Reproducibility:** Everything runs and can be verified
- **Insight:** Novel findings or meaningful contributions

### Think about it!

Would you be proud to show this project in a job interview or grad school application? Would you be comfortable presenting it at a research conference? That's the bar we're aiming for.

# Today's Work Session

## Use today for:

- **Debugging:** Fix critical bugs, resolve issues
- **Completing experiments:** Finish any remaining analyses
- **Creating presentation:** Draft slides, record video
- **Testing reproducibility:** Run notebook in clean Colab
- **Peer feedback:** Show work to other teams, get suggestions
- **Getting help:** Ask questions, troubleshoot with instructor

## Office Hours Available

I'll be available all day for questions, debugging help, and feedback. Use this time!

## Pro Tip

Work together in your team, but also talk to other teams. Fresh perspectives often reveal issues you've been missing!



# Getting Unstuck

## If you're stuck on a technical issue:

- 1 **Read the error message carefully:** What exactly is breaking?
- 2 **Google the error:** Someone has likely encountered it before
- 3 **Check documentation:** Are you using the API correctly?
- 4 **Simplify:** Create minimal example that reproduces the issue
- 5 **Use GenAI tools:** Claude, ChatGPT can help debug
- 6 **Ask for help:** Discord, peers, instructor

## If you're stuck on project direction:

- 1 **Review your proposal:** What was your original plan?
- 2 **Prioritize:** What's essential vs. nice-to-have?
- 3 **Scope down:** Better to do less well than more poorly
- 4 **Talk it through:** Explain the issue to a teammate or peer
- 5 **Consult instructor:** Get feedback on revised approach

# Key Takeaways

- 1 **Prioritize ruthlessly:** Focus on what matters most
- 2 **Test reproducibility:** Run in clean Colab before submitting
- 3 **Be honest about limitations:** Negative results are OK
- 4 **Practice your presentation:** Time it, refine it, polish it
- 5 **Use available support:** Office hours, peers, Discord
- 6 **Start with working code:** Polish can come later
- 7 **Document as you go:** Don't leave it all for the last minute

## Remember

This is the culmination of everything you've learned. From ELIZA to GPT, from pattern matching to transformers—you've built an incredible foundation. Now show us what you can do with it!

# Final Submission Checklist

## Before you submit Friday night, verify:

- ☐ Code runs completely in fresh Colab (tested!)
- ☐ Video uploaded to YouTube, link included
- ☐ Writeup is 2-5 pages with all required sections
- ☐ All team members listed on all deliverables
- ☐ References properly cited
- ☐ Presentation ready for Wednesday or Friday
- ☐ All files submitted to Canvas

### Pro Tip

Submit early if possible! Don't wait until 11:59 PM—technical issues happen.

# You've Got This!

You've learned an incredible amount this quarter.

You started with simple string manipulation and pattern matching...

And now you're building sophisticated LLM systems!

**Trust your skills. Use your resources. Do great work.**

We can't wait to see your presentations!

**Now: Let's get to work!**